

Sequence Operations

Example 1 : Sequence Length

The cardinality operator gives the length of a sequence:

$\# \langle \rangle$
 $\# \langle a, b, c \rangle$
 $\# s$

Example 2 : Pattern Matching for Head and Tail

In Z, decomposing a sequence uses pattern matching with concatenation.

Given a non-empty sequence s , if we have x and t such that:

$$\langle x \rangle \frown t = s$$

Then x is the first element (head) and t is the remaining sequence (tail).

Example 3 : User - Defined First and Rest Functions

You can define explicit functions for sequence decomposition:

$[X]$

$$\left| \begin{array}{l} fst : seq_1 X \rightarrow X \\ rest : seq_1 X \rightarrow seq X \end{array} \right| \begin{array}{l} \forall x : X; s : seq X \bullet fst(\langle x \rangle \frown s) = x \\ \forall x : X; s : seq X \bullet rest(\langle x \rangle \frown s) = s \end{array}$$

Example 4 : User - Defined Last and Init Functions

$$\left| \begin{array}{l} lst : seq_1 X \rightarrow X \\ init : seq_1 X \rightarrow seq X \end{array} \right| \begin{array}{l} \forall x : X; s : seq X \bullet lst(s \frown \langle x \rangle) = x \\ \forall x : X; s : seq X \bullet init(s \frown \langle x \rangle) = s \end{array}$$

Example 5 : Domain and Range

Sequences are functions from indices to elements:

$\text{dom} \langle \rangle$
 $\text{dom} \langle a, b, c, d \rangle$
 $\text{ran} \langle a, b, c \rangle$

Example 6 : Concatenation

The concatenation operator joins sequences:

$s \frown t$
 $\langle 1, 2 \rangle \frown \langle 3, 4 \rangle$
 $\langle a \rangle \frown \langle b, c \rangle \frown \langle d \rangle$

Example 7 : Using Pattern Matching

Pattern matching extracts elements from sequences:

$$\forall s : \text{seq}_1 \mathbb{N} \bullet \exists x : \mathbb{N}; t : \text{seq } \mathbb{N} \bullet s = \langle x \rangle \frown t$$

This asserts that every non-empty sequence can be decomposed.

Example 8 : Recursive Function Example

Define a sum function using pattern matching style:

$$\left| \begin{array}{l} \text{sumSeq} : \text{seq } \mathbb{N} \longrightarrow \mathbb{N} \\ \hline \text{sumSeq}(\langle \rangle) = 0 \\ \forall x : \mathbb{N}; s : \text{seq } \mathbb{N} \bullet \text{sumSeq}(\langle x \rangle \frown s) = x + \text{sumSeq}(s) \end{array} \right.$$